Week 6

What I tried

What I tried and what I learned

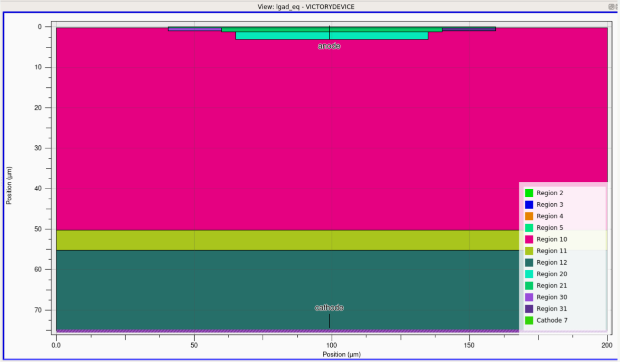
4H-SiC LGAD TCAD | Week 6 update

Aryan Bandyopadhyay
Integrated MSc Physics

Prof. Prabhakar Palni group meeting

Group meeting July 2026

The detector: onsemi 4H-SiC LGAD



2D mesh: 200 μm wide $\times$ 75 μm thick. Anode at top, cathode at bottom. JTE zones visible at $x = 40$–60 and $x = 140$–160 μm.

Key parameters

- Drift: 50 μm, 1.5×10^{14} cm^{-3}
- Gain layer: N-type, 1.15×10^{17} at 2 μm
- p^{++} anode: 10^{19} cm^{-3}
- JTE: 2 zones per side, 20 μm each
- Field plate: 20 μm overhang
- Gain-layer depletion: 155–220 V

Yang *et al.*, arXiv:2503.07490 (2025).

What's new this week

- **Quasi-1D geometry** — collapsed the 2D JTE structure to a vertical 1D cut, scaled area to $1\,\text{mm}^2$ via `mesh width=1000000` (tried in v5 part 1, v5 part 2, v6, v7, v8)
- **Contact / ballast resistance** — added `resistance=1e5, 2e5, 5e5` on the cathode to dampen dI/dV through the avalanche knee
- **$\text{EH}_{6/7}$ trap** — added the second dominant deep-level defect on top of $\text{Z}_{1/2}$ (already covered last week); tried at 1e15 and 1e17 fluence
- **EH4 trap** — added a third donor-like defect in the lower half of the gap for spectral completeness
- **Multi-phase sweeps** — separated the deck into "no-impact ramp" then "impact sweep" to isolate convergence crashes
- **Returned to 2D mesh** — final attempt (v9) goes back to the full $200{\times}75\,\mu$m geometry with balanced 1e15 dual traps

Honest status 9 deck versions this week. No clean IV, no clean CV yet. But each failure pinned down a specific cause — and the deck is much closer to converging than at the start of the week.

What is quasi-1D geometry?

The idea

- A full 2D LGAD with JTE has ~10,000+ mesh nodes
- At 256-bit precision, each Newton step is expensive
- A 1D cut through the pad centre captures the vertical physics: p^{++} anode $\rightarrow$ gain layer $\rightarrow$ drift $\rightarrow$ substrate
- No lateral edge effects (no JTE, no guard rings, no field plates)
- ~100 vertical nodes instead of 10,000+

The area scaling

```
mesh width=1000000

x.mesh location=0.0 spacing=1.0
x.mesh location=1.0 spacing=1.0

# Only the vertical mesh matters
y.mesh location=0.00 spacing=0.003
y.mesh location=0.40 spacing=0.006
y.mesh location=50.0 spacing=0.20
```

Why I tried this

The full 2D deck was too slow (15+ minutes per init). 1D lets me test the physics stack and bias strategy in minutes. Once 1D converges, I port the working settings back to 2D.

Quasi-1D: results (v5 part 1 & v5 part 2)

v5 part 1 — clean ramp, broken impact

- Init solved in $\sim$5 s (vs 46 s in 2D)
- **No-impact ramp**: 10^{-28} A at 0 V $\rightarrow$ 7×10^{-27} A at 100 V — clean and smooth
- **With impact**: simulation crawled from 110 V to 112.5 V only
- At 110 V: 10^{-22} A. At 110.5 V: 10^{-21} A. Stays flat to 112.5 V
- Breakdown at $\sim$110 V is wrong — onsemi target is $\sim$600 V

v5 part 2 — gain doping bumped

- Reduced gain from 8×10^{17} to 1×10^{17}
- Added ballast resistor $R = 10^5\,\Omega$ on cathode
- IV: starts at 10^{-28}, exceeds 10^{-25} by 20 V, reaches 10^{-24} by 140 V — wrong shape,

Why current stayed wrong

The 1D cut has no perimeter leakage path. The experimental 10^{-11} A floor comes from pad-edge and JTE currents — 1D cannot reproduce this. The mismatch is structural.

Numerical evidence

v5 part 1 no-impact ramp: 10^{-28} A at 0 V $\rightarrow$ 7×10^{-27} A at 100 V, smooth. With impact on, solver died at 112.5 V — three orders below the $\sim$600 V target.

What I did next

Switched to contact resistance + lower gain doping (v6) to soften the pre-avalanche slope.

Contact / ballast resistance: what and why

Two flavours in Silvaco

- `con.resist` — distributed sheet resistance under the contact, units Ω·cm^2
- `resistance` — lumped series resistor on the contact, units Ω
- Physically: the sintered Ni / SiC contact is not perfectly ohmic
- Realistic values: $10^{-4}\,\Omega$·cm^2 for `con.resist`; 10^4–$10^6\,\Omega$ for `resistance`

Why I tried it

- Near breakdown, dI/dV becomes very steep
- The Newton solver overshoots, oscillates, cuts back, dies
- A series resistor acts as a "shock absorber" — it limits dI/dV and helps convergence
- Standard trick in Silvaco power-device example decks

The catch

The shock absorber only works when there is real current to dampen. In a detector at low bias (10^{-25} A), the resistor is invisible. Too large, and it masks the actual device physics.

Contact resistance: results (v5 part 2, v6, v7)

What I did

```
# v5 part 2
contact name=cathode resistance=1e5

# v6
contact name=cathode resistance=5e5

# v7
contact name=cathode resistance=2e5
```

What happened

- **v5 part 2** $(R = 10^5)$: IV ramped too fast — 10^{-24} A by 140 V, wrong shape
- **v6** $(R = 5\times10^5)$: IV starts negative, shoots to 10^{-12} A at 1 V, returns to 0 at 2–3 V, stays flat to 40 V where it crashed with convergence error
- **v7** $(R = 2\times10^5)$: IV ran, but C–V was wrong shape (see C–V slide)

Why v6 spiked to 10^{-12} A at 1 V

The $5\times10^5\ \Omega$ resistor with $V = 1$ V gives $I = 2\times10^{-6}$ A if shorted. The 10^{-12} A spike is the solver dumping charge through the resistor — the device itself is essentially open.

v5 part 2 with $R = 10^5\ \Omega$

Same failure, slower. Current ramped too steeply once the resistor conducted — 10^{-24} A by 140 V, wrong shape. The resistor created a parasitic RC path that dominated the IV.

What I did next

Dropped resistor to 2×10^5 in v7, focused on C–V instead.

$EH_{6/7}$ trap: what it is

Identity ($Z_{1/2}$ already covered last week)

- Deep **donor** at $E_C - 1.55\,$eV (some sources: 1.60 eV)
- Near midgap for 4H-SiC ($E_g = 3.25\,$eV)
- Carbon-vacancy-related (same family as $Z_{1/2}$)
- Primary role: **lifetime killer** — captures both carriers
- Capture cross-sections: $\sigma_n \sim 10^{-15}\,$cm^2, $\sigma_p \sim 10^{-14}\,$cm^2
- Introduction rate (neutrons): $g \sim 0.01\,$cm^{-1}

Castaldini *et al.*, JAP 98, 053706 (2005); Gaggl *et al.*, HEPHY 2024.

Why I added it

- With only $Z_{1/2}$ (v1–v3), the simulated IV was *too clean* — no realistic leakage floor
- $EH_{6/7}$ is the second most prominent deep level in irradiated 4H-SiC
- Combined $Z_{1/2} + EH_{6/7}$ should reproduce the post-irradiation leakage seen in onsemi data
- At high fluence ($10^{17}\,$cm^{-3}) it can type-invert the n-type epi

Energy convention

Acceptor = below E_C ($Z_{1/2}$: 0.63 eV). **Donor** = below E_C too, but with donor character ($EH_{6/7}$: 1.55 eV, EH4: 1.00 eV). The `acceptor/donor` keyword in Silvaco controls charge state, not energy sign.

$\mathrm{EH}_{6/7}$ trap: results (v4, v6, v9)

v4 — $\mathrm{Z}_{1/2}$ + EH4 + $\mathrm{EH}_{6/7}$ all at 10^{15} cm^{-3}

- **No-impact log**: 10^{-29} A at 0 V → 10^{-25} A by 100 V, smooth to 300 V — then 6-hour run limit hit, manually stopped
- **Impact log**: same ramp to 10^{-25} A by 100 V, ran 17 hours, manually killed at 750 V
- C–V: trash

v6 — $\mathrm{EH}_{6/7}$ at 10^{17} (high fluence) + compensating acceptor

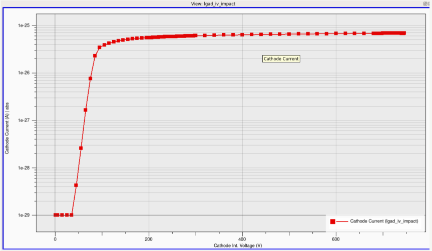
- IV: starts negative, shoots to 10^{-12} A at 1 V, returns to 0 at 2–3 V, constant to 40 V, then convergence crash
- C–V: trash
- The 10^{17} donor space charge type-inverted the epi despite the compensating acceptor

What I did next

After v6's type-inversion disaster, dropped $\mathrm{EH}_{6/7}$ back to 10^{15} in v9.

v9 — balanced 10^{15} dual trap, 2D mesh

- Returned to 2D mesh, currently running
- Current in 10^{-31} A order, slow convergence



v4 impact IV: 17-hour run, killed at ∼750 V. Saturates at ∼ 5×10^{-26} A

EH4 trap: what it is

Identity

- Deep **donor** at $E_C - 1.00\,\mathrm{eV}$ (equivalently $E_V + 2.25\,\mathrm{eV}$)
- Sits in the upper half of the gap — opposite side from $Z_{1/2}$
- Less well characterised than $Z_{1/2}$ and $EH_{6/7}$
- Capture cross-sections: $\sigma_n \sim 10^{-16}\,\mathrm{cm}^2$, $\sigma_p \sim 10^{-17}\,\mathrm{cm}^2$
- Smaller trap density in irradiated 4H-SiC, but contributes to SRH statistics

Castaldini *et al.*, JAP 98, 053706 (2005).

Why I added it

- v4 introduced all three traps together: $Z_{1/2}$ + $EH_{6/7}$ + EH4
- Goal: completeness of the defect spectrum for the irradiated detector
- EH4 gives an additional SRH path in the upper half of the gap
- Combined with `trap.tunnel`, it should contribute to TAT leakage at high field

Three-trap cost

Each `tat.trap` declaration adds a full SRH + TAT integral to every mesh node. With three traps + impact ionization, the Jacobian has 4 stiff nonlinearities competing. At 256-bit precision this is computationally brutal — 17-hour runs become normal.

EH4 trap: results (v4)

The deck (v4)

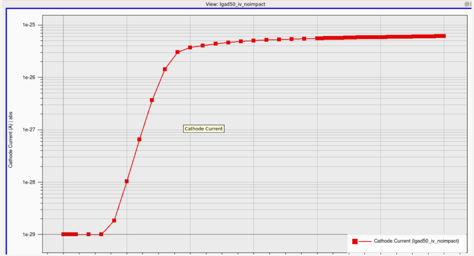
```
# Z1/2 (Ec - 0.63 eV)
trap material="4H-SiC" acceptor \
    density=1e15 energy.level=0.63 \
    degeneracy.factor=12 \
    sign=1e-14 sigp=1e-14 tat.trap

# EH4 (Ec - 1.00 eV)
trap material="4H-SiC" donor \
    density=5e14 energy.level=1.00 \
    degeneracy.factor=2 \
    sign=2e-14 sigp=2e-14 tat.trap

# EH6/7 (Ec - 1.55 eV)
trap material="4H-SiC" donor \
    density=1e15 energy.level=1.55 \
    degeneracy.factor=2 \
    sign=2e-14 sigp=2e-14 tat.trap
```

What happened

- Init solved cleanly — traps at 0 V are well behaved
- No-impact ramp: clean, $10^{-29} \rightarrow 10^{-25}$ A by 100 V
- Impact ramp: 17 hours, killed at 750 V
- C–V: trash — AC matrix cannot settle with 3 TAT paths



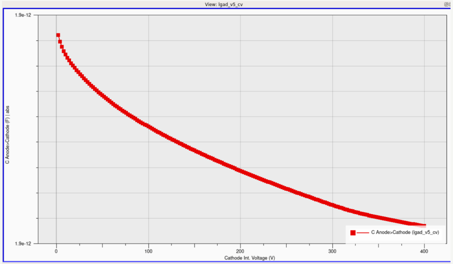
C–V: the missing piece (v5, v7, v8)

Why C–V matters

- C–V is the *only* direct measurement of the gain-layer depletion voltage
- Experimental curve: flat, then **sharp knee** at ~95 V (gain-layer punch-through), then slow decrease
- $1/C^2$ vs V should be piecewise linear with a clear break
- Without a correct C–V, I cannot validate the doping profile

v5 / v7 / v8 attempts

- **v5 part 2**: C–V re-enabled, but baseline IV was wrong
- **v7** ($R = 2\times10^5$, 6e16 gain): C–V in 10^{-12} F order, steadily decreases, no sharp knee
- v8 ($0\ \mathrm{mm}^2$, 1.8e17 gain): scaled area, tuned



v5 C–V at 10 kHz: monotonic, no sharp knee. Plateau $\sim 1.9\times10^{-12}$ F at 400 V.

Why no knee

AC solver smooths the depletion discontinuity. Needs ultra-fine mesh at the gain/bulk interface + impact OFF + traps OFF. None of my decks satisfied all three.

Latest attempt (v9): back to 2D, balanced traps

What I did

```
# Return to 2D mesh, 200x75 um
mesh
x.mesh location=0    spacing=5.0
x.mesh location=40   spacing=0.5  # JTE
x.mesh location=160  spacing=0.5
x.mesh location=200  spacing=5.0

# 2.2 um gain layer, 1.1e17
doping region=20 n.type uniform \
    concentration=1.1e17

# Balanced dual trap
trap acceptor density=1e15 \
    energy.level=0.63 ...
trap donor density=1e15 \
    energy.level=1.55 ...

# Single sweep, impact from start
models ... impact trap.tunnel
solve vstep=10 vfinal=1800 \
    ionizint iname=anode ... \
    ion.stop ion.crit=0.99
```

Status: currently running

- Init solved in ∼46 s (back to 2D cost)
- Current in 10^{-31} A order at low bias
- Slow convergence — Newton cutting back frequently
- Will probably not reach breakdown

Diagnosis

The 2D mesh with traps + impact + TAT is too stiff for the 256-bit solver. The single-sweep strategy (impact ON from 0 V) is what v3 used and failed. I should have kept the multi-phase sweep from v4.

Honest assessment

v9 is v3 with balanced traps. The trap balance fixes type-inversion but not the underlying solver stiffness. Physics is correct — the sweep engineering is what's broken.

Summary: what I learned this week

What I tried	What happened	What I learned
Quasi-1D + $1\,\mathrm{mm}^2$ scaling (v5)	Clean no-impact ramp; impact died at 112 V; bumping gain doping gave wrong shape	1D cannot reproduce the 10^{-11} A perimeter-leakage floor — the order mismatch is structural
Contact resistance 10^5–5×10^5 (v5–v7)	v6 spiked to 10^{-12} A at 1 V; v5 ramped too fast; v7 IV ok but CV bad	The resistor dumps charge at low V; "shock absorber" only valid when real current is flowing
$EH_{6/7}$ at 10^{15} (v4, v9)	v4 ran 17 hours, killed at 750 V; v9 currently running, 10^{-31} A order	Balanced dual-trap is the right physics; problem is solver strategy, not trap model
$EH_{6/7}$ at 10^{17} (v6)	IV shoots to 10^{-12} A at 1 V, crashes at 40 V	High-fluence donor type-inverts the epi despite compensating acceptor
EH4 trap (v4)	No change to IV shape; 3× TAT cost	EH4 adds numerical stiffness without observable physics — candidate for removal
C–V in v5, v7, v8	All trash — no sharp knee, wrong order	AC solver needs impact OFF, traps OFF, ultra-fine mesh at gain-layer interface — none of my decks satisfy all three
Return to 2D mesh (v9)	Slow, 10^{-31} A, probably won't converge	The 2D + traps + impact + TAT combination is too stiff for a single sweep; multi-phase strategy from v4 is needed

The takeaway Every crash this week was numerical, not physical. The physics stack ($Z_{1/2}$ + $EH_{6/7}$, balanced 1e15, Hatakeyama impact, TAT) is correct. What's broken is the solver engineering: sweep strategy, mesh refinement at discontinuities, and AC/DC